

Beamer⁺

Interactive Presentations

Chandra Gummaluru

`chandra-gummaluru.github.io`

The Team

Hasan Malik

Clara Fan

Xiaowu Wu

Chenhao Zhou

what's inside

Beamer⁺ turns a static PDF into a live, interactive lecture
every tile below has a demo later in this deck — press → to explore



video & audio



3D models



split view



reveal steps



ink & shapes



magnifier



live widgets

```
>>>_
```

Python



live polls



viewer sync



screen recording

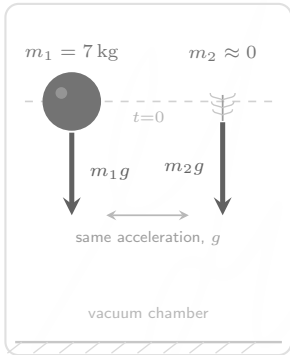


notes PDF

videos/audio

supported formats include .mp4, .mov, .mpeg, .mp3, .wav

example usage: demonstrating Newton's 2nd law in a Physics course



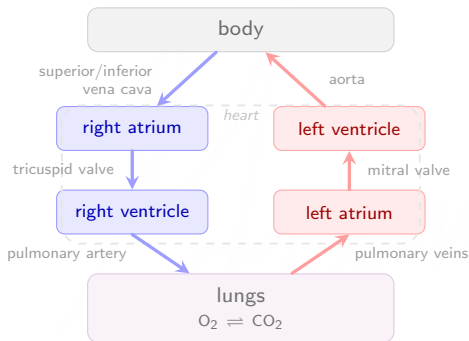
gravitational acceleration is
independent of mass



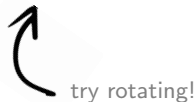
models

supported formats include .glpb

example usage: explaining blood flow through the heart in an Anatomy course



blood flow through the heart
(deoxygenated blood / oxygenated blood)



reveal steps

uncover a slide one piece at a time — no \pause or recompiling
example usage: walking through a proof in a Discrete Math class

Claim. $\sqrt{2}$ is irrational.

- 1 Suppose not: $\sqrt{2} = \frac{p}{q}$ with p, q integers in *lowest terms*.
- 2 Squaring, $p^2 = 2q^2$, so p^2 is even $\Rightarrow p$ is even: $p = 2k$.
- 3 Then $4k^2 = 2q^2$, so $q^2 = 2k^2 \Rightarrow q$ is even too.
- 4 Both p and q are even — they share a factor of 2.

Contradiction! Lowest terms can't both be even. ■

 press → to reveal!

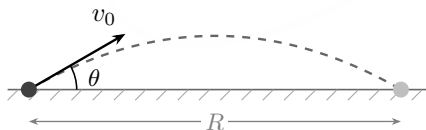
ink & annotations

pens, highlighters, shapes, $\text{\LaTeX}$ text boxes, a spotlight and a magnifier
example usage: working a projectile problem live in a Physics class

A ball is kicked at $v_0 = 20 \text{ m/s}$, at $\theta = 30^\circ$ above level ground. How far away does it land?

The range is

$$R = \frac{v_0^2 \sin 2\theta}{g} = \frac{(20)^2 \sin 60^\circ}{9.81}$$



psst — you found the fine print! The magnifier zooms 1.5 $\times$ to 5 $\times$; scroll while it's active to change the zoom. It even magnifies live video, ink and widgets.

 try the magnifier on the fine print!

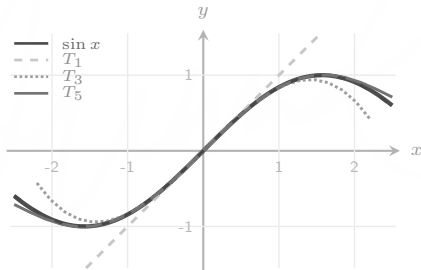
split view

allows two slides to be shown simultaneously

example usage: explaining Taylor series in a Calculus class

Any smooth function f can be approximated around $x_0 = 0$ as:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$



expansion of $f(x) = \sin(x)$ around $x_0 = 0$

function plotter

2D curves and 3D surfaces you can rotate, zoom and edit live
example usage: classifying critical points in a Multivariable Calculus class

At a critical point, $\nabla f = \mathbf{0}$. The
second-derivative test looks at

$$D = f_{xx}f_{yy} - f_{xy}^2.$$

For $f(x, y) = ax^2 - y^2$ at the origin:

$$D = (2a)(-2) - 0^2 = -4a$$

$a > 0$: a **saddle point**

$a < 0$: a local maximum

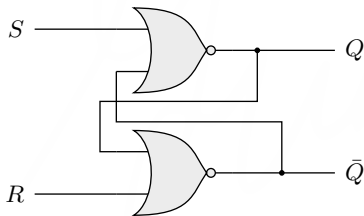
 try dragging the slider for a !

widgets

supported formats include `.html`

example usage: explaining latches in a Computer Organization class

A **latch** stores one bit of state using cross-coupled gate feedback.



cross-coupled NOR gates
inputs are *active-high*

S	R	Q	$\bar{Q}$	state
0	0	Q	$\bar{Q}$	hold
1	0	1	0	set
0	1	0	1	reset
1	1	0	0	forbidden

truth table for an SR latch
implemented with NOR gates

digital logic

truth tables, Karnaugh maps and binary arithmetic, step by step

example usage: simplifying a majority function in a Digital Logic class

The *majority* function is 1 when
at least two of its inputs are 1:

<i>A</i>	<i>B</i>	<i>C</i>	<i>F</i>
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Grouping the 1s on
the map gives

$$F = AB + BC + AC$$

 try stepping through the groups!

record & share

the lecture doesn't have to stay in the room

example usage: posting the recording and annotated notes right after class



record the tab

slides, ink, widgets, mic
and sound — saved
every second, so a
crash loses nothing



notes PDF

one handout with
your ink, every wid-
get as it stands,
and the poll results



viewer page

the audience follows
along on their own
screens, synced live

also: bookmarks · hide the slide with . · save everything as one ZIP · works offline as an app

try the record button!



learn more

visit the link below to learn more

`chandra-gummaluru.github.io/post/beamer-plus`